

E.1 / PAPER 2 / QUESTION 1

26 October 2023 / TZ1 (file label) / HL Paper 2 / original Q5 / 8823-6508

1. The table shows some of the energy levels for a hydrogen atom.

	Level	Energy/eV
Ionized state	$n = \infty$	0
Excited states	$n = 3$	-1.51
	$n = 2$	-3.40
Ground state	$n = 1$	-13.6

- (a) Distinguish between an atom in an excited state and an atom in an ionized state.

[1]

- (b) Explain how electromagnetic radiation is emitted from a hydrogen atom in an excited state.

[2]

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(Question 1 continued)

- (c) When radiation of wavelength $6.6 \times 10^{-7} \text{ m}$ is incident on cold hydrogen gas some of it is absorbed by the gas.

- (i) State the region of the electromagnetic spectrum of this radiation. [1]

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- (ii) Determine the initial and final states of the hydrogen atom that are involved in this absorption. [3]

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